MARCELO VENTURINI

Biomedical Engineer

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PROFILE

Biomedical Engineer with an electronics-oriented background and hands-on experience in biomedical instrumentation, embedded systems, physiological signal acquisition, electronic measurements, and signal processing. Experience developing sensor-based systems with microcontrollers and analog electronics, with practical laboratory experience in measurement and hardware debugging. Interested in R&D roles across medical devices, electronics, embedded systems, and health technology.

EDUCATION

M.Sc. in Biomedical Engineering, Politecnico di Torino, Italy **109/110** 2024–2026
Thesis: *EMG Signal Deconvolution as Preprocessing for Enhanced Hand Gesture Recognition.*

B.Sc. in Biomedical Engineering, Universidad Nacional de Córdoba, Argentina **8.54/10** 2020–2024
Competitive scholarship for international study at Politecnico di Torino. Relevant coursework: Digital Electronics, Biomedical Instrumentation, Signal Processing, Embedded Systems.

TECHNICAL SKILLS

Test & Lab	Oscilloscopes, multimeters, signal generators, soldering, hardware debugging
Programming	C++, Python, Assembler, MATLAB
Software & Tools	Docker, Simulink, Multisim, Arduino IDE, Excel
Embedded & HW	ESP32, PIC microcontrollers, CAN bus, Sensor interfacing, Analog and Digital Circuit design
Languages	Spanish (native), English (C1), Italian (fluent)

RELEVANT EXPERIENCE

Teaching Assistant – Digital Electronics 2024
Universidad Nacional de Córdoba

- Supported undergraduate laboratory sessions and practical exercises in digital electronics.
- Assisted students with logic circuits, measurements, debugging, and experimental results.

SELECTED PROJECTS

M.Sc. Thesis – EMG Gesture Recognition *Politecnico di Torino*

- Developed a real-time sEMG acquisition and processing pipeline using the OT Bioelettronica Sessantaquattro+ and a custom MATLAB interface.
- Implemented digital filtering, EMG signal deconvolution, feature extraction and selection, and evaluated LDA and nonlinear classification methods for online gesture recognition.

Levodopa Response Assessment Device *Incubated Project*

- Designed an analog circuit for sEMG signal acquisition and conditioning and programmed an ESP32 for real-time data acquisition.
- Evaluated Parkinson's treatment response through quantitative physiological measures.

Ultrasonic Sensor Interfacing with PIC *Academic Project*

- Programmed a PIC microcontroller in assembler for timing-sensitive ultrasonic acquisition, with direct hardware interfacing and low-level debugging.

IoT Health & Home Monitoring Platform *Academic Project*

- Integrated HR, SpO_2, and environmental sensors via MQTT using the SenML standard.
- Developed a RESTful CherryPy backend for device and data management, with Telegram and ThingSpeak integration.

Additional Experience
Sailing Coach 2022–2026
Hampton YC (USA), Cold Spring Harbor (USA), Huguenot YC (USA), Orza Minore (Italy)

Further information on projects, technical documentation and certifications: marcelo-venturini.github.io